

Secure E-Commerce Website Project Proposal

Digital Art & Prints Store

Introduction

For this project, we will design a secure e-commerce website focused on selling digital art and print products. The store will mostly deliver downloadable content, which introduces unique security concerns in comparison to physical product stores.

Because digital files can be copied and accessed without authorization, security will be the main focus from the beginning. Our goal is to design a functional e-commerce website while integrating strong authentication methods and a secure payment gateway to protect users and transactions.

A. E-Commerce Selection

We selected a digital art & prints store as our project model. The store will sell:

- Digital art downloads (PNG and JPG files)
- Printable wall art
- Poster prints
- Canvas prints

This option works well because it allows us to focus on both transaction security and secure file delivery. Unlike physical goods, digital products require additional access controls to prevent unauthorized downloads.

B. Project Scope

Our e-commerce website will include the following features:

Product Catalog

- Display artwork with previews
- Product descriptions and pricing
- Category filtering
- Search functionality

User Accounts

- Account creation and login
- Email verification
- Password reset
- Multifactor authentication (MFA)

Shopping Cart & Checkout

- Add and remove products
- Order total calculation
- Secure checkout process
- Confirmation email after purchase

Digital File Delivery

- Secure download links
- Access only after successful payment
- Time-limited download access

Admin Panel

- Upload and manage products
- View orders
- Role-based access control

The website will be designed for ease of use, but security will remain a priority throughout development.

C. Security Challenges

E-commerce websites face several common security challenges:

1. Unauthorized Access to Digital Files

If file storage is not properly secured, users may bypass payment and access downloads directly.

2. Account Compromise

Weak login systems can lead to account takeovers through phishing or brute-force attacks.

3. Payment Fraud

Credit card fraud and transaction manipulation are significant risks in online platforms.

4. Data Breaches

Customer information such as emails and passwords must be protected from exposure.

5. Web Application Attacks

Threats such as SQL injection and cross-site scripting can compromise the system if not properly addressed.

These risks require a layered security approach rather than relying on a single control.

D. Security Solutions

To address these challenges, we will implement the following solutions:

Multifactor Authentication (MFA)

Users will be required to verify their identity using a second factor during login, such as:

- A one-time passcode sent via email
- An authenticator app code

MFA significantly reduces the risk of unauthorized access even if passwords are compromised.

Secure Payment Gateway

We will integrate a trusted third-party payment processor instead of storing credit card information on our servers. This ensures:

- Encrypted transactions
- Reduced risk of storing sensitive payment data
- Compliance with security standards

All transactions will occur over HTTPS using TLS encryption.

Secure File Access Controls

- Digital files will not be publicly accessible
- Download links will be generated after payment
- Links will expire after a set time
- Access will require user authentication

This prevents simple link sharing and unauthorized downloads.

Application Security Controls

- Password hashing using secure algorithms
- Input validation to prevent SQL injection
- Protection against cross-site scripting
- Secure session management

E. Risk Assessment

Even with MFA and a secure payment gateway, some risks may remain.

Potential Risks

- Phishing attempts targeting users
- Insider misuse of administrative privileges
- Vulnerabilities in third-party libraries
- Users redistributing purchased digital content

Mitigation Strategies

- Regular and ongoing software updates and patch management
- Continuous monitoring and logging of suspicious activity
- Least privilege access for administrators
- Limiting download attempts per purchase

By using layered security controls, we reduce the overall risk and strengthen system resilience.

F. Project Timeline

Weeks 1–3:

Finalize requirements and design system architecture.

Weeks 4–6:

Develop front end layout and product catalog. Implement user registration.

Weeks 7–9:

Implement MFA and integrate a secure payment gateway. Develop secure download functionality.

Weeks 10–11:

Conduct testing, including authentication and input validation testing.

Weeks 12–13:

Complete documentation and final revisions.

Week 14:

Submit and present the final project.

Conclusion

Security will be built into the design from the beginning rather than added afterward. Through layered defenses and structured risk management, this project demonstrates how digital content can be delivered securely in an online environment.